

Inner Products and Eigenfunctions in Function Spaces

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Inner Product

An inner product is an operation that combines two vectors in $\mathbb{R}^n$ to produce a single real number (a scalar).

If

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

then the inner product is:

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Key Points:

- The inner product is often called the dot product in $\mathbb{R}^n$.
- Vectors $\mathbf{u}$ and $\mathbf{v}$ are treated as $n \times 1$ matrices.
- The inner product is the matrix multiplication $\mathbf{u}^T \mathbf{v}$, resulting in a 1×1 matrix (a scalar).
- It measures how much two vectors "line up" with each other.
- Allows us to define:
 - Length of a vector
 - Angle between vectors
 - Orthogonality of vectors

Inner Product on a Vector Space

An **inner product** on a vector space V is a function that, to each pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in V , associates a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ and satisfies the following axioms:

- 1 $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
- 2 $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
- 3 $\langle c\mathbf{u}, \mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$
- 4 $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$, and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ iff $\mathbf{u} = \mathbf{0}$

A vector space with an inner product is called an **inner product space**.

Inner Product of Integral

(For functions $f(x), g(x), h(x)$ over an interval $[a, b]$)

The inner product is defined by:

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx$$

Properties of Integral Inner Product

(a) Symmetry

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx = \int_a^b g(x)f(x) dx = \langle g, f \rangle$$

(b) Additivity

$$\begin{aligned}\langle f + g, h \rangle &= \int_a^b (f(x) + g(x))h(x) dx \\ &= \int_a^b f(x)h(x) dx + \int_a^b g(x)h(x) dx \\ &= \langle f, h \rangle + \langle g, h \rangle\end{aligned}$$

Properties of Integral Inner Product

(c) Scalar Multiplication

$$\begin{aligned}\langle cf, g \rangle &= \int_a^b (cf(x))g(x) dx \\ &= c \int_a^b f(x)g(x) dx \\ &= c\langle f, g \rangle\end{aligned}$$

(d) Positivity

$$\langle f, f \rangle = \int_a^b f(x)^2 dx \geq 0$$

■ $f(x)^2 \geq 0$ for all real x : $\langle f, f \rangle = 0$ if and only if $f(x) = 0$

Sum Inner Product for Vectors in $\mathbb{R}^n$

(For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$)

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i v_i = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Properties of Sum Inner Product

(a) Symmetry

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

$$= v_1 u_1 + v_2 u_2 + \cdots + v_n u_n$$

$$= \langle \mathbf{v}, \mathbf{u} \rangle$$

Properties of Sum Inner Product

(b) Additivity

$$\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \sum_{i=1}^n (u_i + v_i) w_i$$

$$= (u_1 + v_1)w_1 + (u_2 + v_2)w_2 + \cdots + (u_n + v_n)w_n$$

$$= (u_1w_1 + v_1w_1) + (u_2w_2 + v_2w_2) + \cdots + (u_nw_n + v_nw_n)$$

$$= (u_1w_1 + u_2w_2 + \cdots + u_nw_n) + (v_1w_1 + v_2w_2 + \cdots + v_nw_n)$$

$$= \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$$

Properties of Sum Inner Product

(c) Scalar Multiplication

$$\langle c\mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n (cu_i)v_i$$

$$= (cu_1)v_1 + (cu_2)v_2 + \cdots + (cu_n)v_n$$

$$= c(u_1v_1) + c(u_2v_2) + \cdots + c(u_nv_n)$$

$$= c(u_1v_1 + u_2v_2 + \cdots + u_nv_n)$$

$$= c\langle \mathbf{u}, \mathbf{v} \rangle$$

Properties of Sum Inner Product

(d) Positivity

$$\langle \mathbf{u}, \mathbf{u} \rangle = \sum_{i=1}^n u_i^2 = u_1^2 + u_2^2 + \cdots + u_n^2 \geq 0$$

Since each $u_i^2 \geq 0$, the whole sum is non-negative.

If $\langle \mathbf{u}, \mathbf{u} \rangle = 0$, then $u_1 = u_2 = \cdots = u_n = 0$, thus $\mathbf{u} = \mathbf{0}$.

Eigenvector and Eigenvalue

An eigenvector of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that

$$A\mathbf{x} = \lambda\mathbf{x}$$

for some scalar λ .

- The scalar λ is called an **eigenvalue** of A .
- The vector $\mathbf{x}$ is called an **eigenvector** corresponding to λ .

Key Points

- A is an $n \times n$ matrix (square matrix).
- $\mathbf{x}$ is a nonzero vector in $\mathbb{R}^n$.
- λ is a scalar in $\mathbb{R}$.
- The equation $A\mathbf{x} = \lambda\mathbf{x}$ means applying A to $\mathbf{x}$ only stretches, shrinks, or flips $\mathbf{x}$, but does not change its direction.
- Eigenvectors show invariant directions under the transformation by A .
- Eigenvalues show how much the eigenvectors are scaled.

Eigenfunctions

$$L(f) = \lambda f$$

- L is the linear operator
- f is the eigenfunction
- λ is an eigenvalue; specify positive!

Eigenfunction computation in $C[a,b]$

LTI systems

Linear time invariant systems

LTI example

$$H(x(t)) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau)d\tau$$

Function Space $L^2[a, b]$

$L^2[a, b]$ is the space of all integrable squared functions.

$$\int_{-\infty}^{\infty} (|f(x)|)^2 dx < \infty$$

Laplace Operator Eigenfunctions

$$L(f) = \lambda f$$

- specify λ is a positive eigenvalue!

Dirichlet Boundary Conditions

$$y(0) = y(\pi) = 0$$

in the infinite function space $L^2[0, \pi]$

Laplace Operator

$$L(f) = -\frac{d^2 y}{dx^2}$$

Known Relationships [4]

Operator/Eigenfunction Relationship

$$L(y) = \lambda y = -y''$$
$$y'' + \lambda y = 0$$

Characteristic Equation for Second Order Harmonic Differential

$$ar^2 + br + c = 0$$

$$r^2 + \lambda = 0$$

$$\implies (r + \sqrt{\lambda}i)(r - \sqrt{\lambda}i) = 0$$

$$\implies r = \pm \sqrt{\lambda}i$$

Solving for Eigenfunctions and Eigenvalues [1]

General Solution for Second Order Harmonic Differential

$$\begin{aligned}y(t) &= c_1 e^{\phi t} \cos(\mu t) + c_2 e^{\phi t} \sin(\mu t) \text{ for } r = \phi \pm \mu i \\&= c_1 e^{\phi \cdot 0} \cos(\sqrt{\lambda} t) + c_2 e^{\phi \cdot 0} \sin(\sqrt{\lambda} t) \\&= c_1 \cos(\sqrt{\lambda} t) + c_2 \sin(\sqrt{\lambda} t).\end{aligned}$$

Plugging in the Dirichlet Boundary Conditions

$$y(0) = y(\pi) = 0$$

General Eigenfunction and Eigenvalue

$$\begin{aligned}\lambda_n &= n^2 \\y_n(t) &= \sin(n \cdot t)\end{aligned}$$

Orthogonality of Eigenfunctions: Inner Products in Function Spaces [2]

- All eigenfunctions y_n are $\in L^2[0, \pi]$ due to the nature of $\sin(x)$
- We can therefore use the L^2 inner product, defined for $f, g \in L^2$ over region Ω as:

$$\langle f, g \rangle = \int_{\Omega} f(x)g(x) dx$$

Choose the following distinct eigenfunctions (different eigenvalues):

$$y_{n_1} = \sin(n_1 t)$$

$$y_{n_2} = \sin(n_2 t)$$

Recall: Application to Spherical Harmonics [3]

- spherical harmonics are eigenfunctions of the angular component of the three-dimensional Laplace Operator
- any two spherical harmonics of different degrees are orthogonal over the 2-sphere, i.e.

$$\langle Y_n(\zeta), Y_m(\zeta) \rangle = \int_{S^2} Y_n(\zeta) \cdot Y_m(\zeta) dS = 0$$

References

- [1] Paul Dawkins. “Second Order DE’s”. In: *Paul’s Online Notes* (2025).
- [2] Richard Fitzpatrick. “Eigenfunctions of Operators are Orthogonal”. In: *Chem LibreTexts* (2025).
- [3] Maxwell Thum Matt DeCross and Jimin Khim. “Spherical Harmonics”. In: *Brilliant.org* (2025).
- [4] William F. Trench. *Elementary Differential Equations with Boundary Value Problems*. Faculty Authored and Edited Books CDs, 2013.